

Transform each of 55-58 by making the change of variable $j = i - 1$.

5.1.58.
$$\prod_{i=n}^{2n} \frac{n-i+1}{n+i}$$

$$\begin{aligned} j &= i - 1 \\ i &= j + 1 \end{aligned}$$

Lower bound:

$$\begin{aligned} j + 1 &= n \\ j &= n - 1 \end{aligned}$$

Upper bound:

$$\begin{aligned} j + 1 &= 2n \\ j &= 2n - 1 \end{aligned}$$

Change of variable:

$$\begin{aligned} (n - i + 1) &= n - (j + 1) + 1 \\ n - (j + 1) + 1 &= n - j - 1 + 1 = n - j \\ (n + i) &= n + (j + 1) = n + j + 1 \end{aligned}$$

Solution:
$$\prod_{j=n-1}^{2n-1} \frac{n-j}{n+j+1}$$